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Data-driven prediction of the equivalent sand-grain roughness

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Surface roughness affects the near-wall fluid velocity profile and surface drag, and is commonly quantified by the equivalent sand-grain roughness k_s. It is essential to estimate k_s for accurate fluid dynamic problem modeling. While numerous roughness correlation formulas have been proposed to predict k_s in the fully rough regime, most of them are restricted to certain roughness types, with various geometric parameters considered in each case, leading to ongoing disagreements regarding its parameterization and lack of universality. In this study, a Particle Swarm Optimized Backpropagation (PSO-BP) method is proposed to predict k_s based on the selected surface parameters from previous DNS, LES, and experimental results for flow behavior over various surface roughness. The PSO-BP model's ability to predict k_s in the fully rough region is evaluated and compared with both the existing roughness correction formulas as well as the traditional BP model. An optimized polynomial function is also proposed to serve as a 'white box' for predicting k_s. It turns out that the PSO-BP method has better performance in the evaluation metrics compared to other methods, yielding a Mean Absolute Error (MAE) of 0.0390, a Mean Squared Error (MSE) of 0.0026 and a Mean Absolute Percentage Error (MAPE) of 28.12%. This novel approach for estimating k_s has practical applicability and holds promise for improving the precision and efficiency of calculations related to equivalent sand-grain roughness, and thus provides more accurate and effective solutions for CFD and other engineering applications.

Surface roughness significantly impacts flow-related engineering applications, leading to increased skin friction compared to smooth surfaces[1], and results in a downshift of near-wall velocity profile which can be quantified by:

$$U^+ = \frac{1}{\kappa} lny^+ + B - \Delta U^+ \tag{1}$$

where $\kappa \approx 0.4$ is the von Kármán constant, $B \approx 5.0$ is the intercept for the smooth wall case and the roughness function ΔU^+ is related to roughness Reynolds number k_s^+ in the fully rough regime via:

$$\Delta U^+ = \frac{1}{\kappa} lnk_s^+ + B - 8.5 \tag{2}$$

in which $k_s^+ = \frac{k_s U_\tau}{\nu}$ and k_s denotes the equivalent sand-grain roughness. And it is conceptualized as an equivalent roughness height for surfaces that are irregular and non-uniform, yielding the same skin-friction coefficient as the uniform sand-grain of the Nikuradse's in the fully-rough regime[2]. When simulating the near-wall flow using Reynolds-Averaged Navier–Stokes (RANS) method in Computational Fluid Dynamics (CFD), surface roughness effects can be accounted for by introducing the equivalent sand-grain roughness length scale k_s into the boundary condition of turbulence models[3]. However, this approach necessitates the a priori knowledge of the k_s value.

There exist numerous correlations for the equivalent sand-grain roughness k_s. Chan[4] confirmed strong correlation exists between roughness function and mean roughness height k_a as well as streamwise effective slope ES over sinusoidal roughness; Forooghi[5] found that a combination of mean maximum peak-to-trough height k_t, skewness Sk and effective slope ES can lead to satisfactory prediction of k_s over randomly distributed roughness elements; Flack[6] proposed a roughness correlation formula based on the root-mean-square roughness height k_{rms} and skewness (S_k) from various grit blasting surfaces, and Kuwata[7] developed a roughness correlation function also based on k_{rms} and S_k over randomly distributed hemispheres, to name a few. Although previous research

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provided valuable insights into the correlation between surface features and the equivalent sand-grain roughness, various geometrical parameters are considered in each roughness correlation formula. Moreover, these correlation formulas are limited to the particular roughness types they investigated, suggesting a potential lack of universality that might constrain the effectiveness when applied to arbitrary surface roughness. Therefore, it is crucial to establish a universal roughness correlation that leverages essential surface features, minimizing complexity while maintaining applicability to a wide variety of roughness types, and also enable the accurate prediction of k_s.

Despite the difficulty of deriving the relationship in the traditional approach, the machine learning (ML) technique is helpful when dealing with the high-order features problem and mapping the complex relationship between surface parameters to equivalent sand-grain roughness. Jouybari[8] used Deep Neural Network (DNN) and Gaussian Process Regression (GPR) methods to train the model and predict k_s with 17 inputs containing both roughness parameters and their products. Even though they achieved relatively low errors compared with existing roughness correlation formulas, these methods have their shortcomings. For example, only 45 different datasets with 17 inputs considered in their study could lead to overfitting and limit DNN's reliability since it's typically reliant on large datasets. Also, the performance of GPR heavily depends on the choice of kernel function and its hyperparameters, which often require further experimentation or cross-validation, and selecting an inappropriate kernel can lead to poor model performance. Furthermore, it took approximately 75 h to obtain the optimal DNN network and 35 h for the GPR method, which is time-consuming.

The Backpropagation (BP) neural network has its advantages like non-linear mapping ability and simple structure with high efficiency. And Particle Swarm Optimization (PSO) is a global optimization technique inspired by the social behavior of bird flocks, and it can help to find a better set of weights and biases for the BP neural network by exploring the search space more thoroughly and reducing the chance of being stuck in the local minimum. In this research, a Particle Swarm Optimized Backpropagation (PSO-BP) neural network is proposed to attempt to achieve a more universal correlation between the most important features (to be analyzed in the next section) and equivalent sand-grain roughness over various rough surface types, ranging from hemispheres, truncated cones, numerically generated surfaces or replicated from real engineering applications, in the fully rough regime. It turns out the PSO-BP method outperforms selected roughness correlation formulas and traditional BP algorithm as well as the optimized polynomial function, hence providing an effective method for estimating k_s under various types of roughness.

Methods

Surface parameters analysis

According to Flack[9] an ideal model for predicting equivalent sand-grain roughness k_s should be based on readily measurable surface parameters, which could be determined via a simple surface scan, thus avoiding the need for further experiments or DNS simulations. Since it's not possible to take all surface geometric features into consideration when predicting k_s, it's necessary to find a minimal yet representative set of surface parameters that can sufficiently convey the drag-producing characteristics over various roughness types[9].

Chung[10] suggested that a robust correlation between surface topography and equivalent roughness requires the consideration of at least three categories. The first one encompasses surface elevation, which can be represented by parameters such as k_a, k_t, k_{rms}, etc. The second involves the effective slope (ES), the frontal solidity (λ_f) or the roughness density parameter (Λ_s). Finally, the third component is characterized by plan solidity (λ_p), solid volume fraction (ϕ), or the skewness (Sk).

When it comes to representing roughness height, the root-mean-square height k_{rms} gives a measure of the standard deviation of the height distribution[10],which means it takes both the amplitude and distribution of surface asperities into account, providing a more comprehensive statistical measure of the surface texture over the former two roughness height, namely, the average roughness height k_a and the mean maximum peak-to-trough roughness height k_t. As different surface structures can possess the same k_a value and k_t is more sensitive to occasional high peaks or deep valleys, both of which can lead to misinterpretation of the surface characteristics.

And the effective slope ES stands for the mean absolute streamwise gradient of the surface roughness and provides a measure of the steepness of roughness features[9]. It's worth noting that the frontal solidity (λ_f), and roughness density parameter (Λ_s) are closely related to the effective slope ES, with $ES = 2\lambda_f$ and Λ_s is related to the local angle α_h whose absolute value is approximately ES[10]. The skewness Sk reveals whether the surface is dominated by peaks ($Sk > 0$) or pits ($Sk < 0$)[9]. In addition, plan solidity (λ_p), which is defined as the ratio of plan area, and solid volume fraction (ϕ), are also have a strong correlation with Sk. Besides, Bhaganagar and Chau[11] found that the kurtosis Ku, which accounts for the sharpness of the height distribution of a surface, could be effective for characterizing the roughness function from their DNS results. The mathematical expressions for k_{rms}, ES, λ_f, Λ_s, λ_p, Sk and Ku are provided in the Supplementary Materials.

Other than the above analysis, the correlation heatmap can offer useful perspective into the relationships among different parameters, particularly in identifying the most influential factors on a given outcome. Thus, a correlation heatmap is created based on data provided by Jouybari[8] where 45 distinct roughness geometries were considered with 12 surface parameters investigated. The color intensity and numeric values within each cell represent the correlation coefficient between variable pairs.

The geometric parameters shown in Fig. 1 are: mean roughness height k_{avg}, roughness peak-to-trough height k_c, mean maximum peak-to-trough height k_t, root-mean-square roughness height k_{rms}, first-order moment of height fluctuations R_a, inclination angle in i directions inc_i, surface porosity P_o, effective slope in i directions ES_i, skewness Sk and kurtosis Ku, respectively. Among the roughness height related parameters ($k_{avg}, k_c, k_t, k_{rms}, R_a$), k_{rms} shows the strongest correlation with k_s and has a coefficient of 0.85, indicating that as k_{rms} increases, the equivalent sand-grain roughness likely increases significantly. Also, Sk and ES_x both exhibit moderate positive

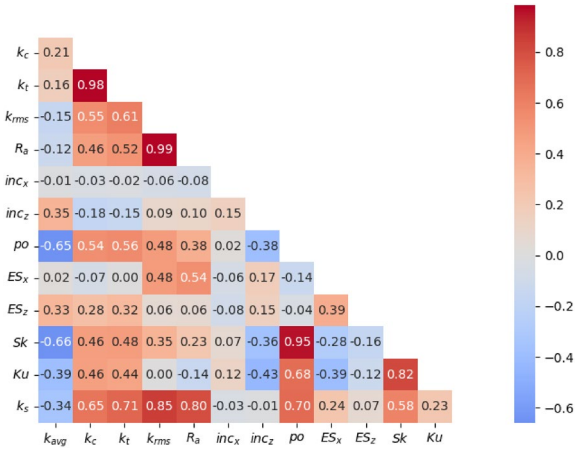

Figure 1. Heatmap of the correlation matrix.

correlations with k_s, coefficients being 0.58 and 0.24 respectively, suggesting that their increase is associated with an increase in k_s, and the streamwise *ES* is more involved than spanwise. Similarly, *Ku* also shows a positive correlation with k_s (coefficient of 0.23). Since the limited data of surface porosity P_o and it also exists strong correlation with *Sk* (0.95), we don't take P_o into further consideration.

The correlation heatmap result substantiates the previous analysis, and hence we adopted the root-mean-square roughness height k_{rms}, streamwise effective slope *ES*, skewness *Sk* and kurtosis *Ku* as the representative surface parameters to serve as the input variables for predicting equivalent sand-grain roughness k_s through the BP and PSO-BP methods.

Particle swarm algorithm and neural network

Backpropagation (BP) is a prominent machine learning algorithm achieved by repeated forward propagation of input and backward propagation of error procedure with an iterative adjustment of weights and biases, until the output error is reduced to an acceptable level or the preset maximum iteration is reached. The BP neural network (BPNN) is well-suited for approximating nonlinear functions with multivariate input features while still maintaining a simple structure. However, the BPNN does come with certain drawbacks, such as the sensitivity to random initial weights and biases, and it's not guaranteed to find the global minimum. Instead, it may get stuck in a local minimum.

And the particle swarm optimization (PSO) is a population-based stochastic optimization technique inspired by the social behavior of bird flocking, where birds adjust their navigation based on individual experience and group communication to help the entire flock get the best hunt[12]. In the PSO technique, each particle is a potential solution to the target problem and has a fitness value which is essentially a score that indicates how well the particle meets the objective function of the optimization problem. And it is obtained from the fitness function by the Root Mean Squared Error (RMSE) calculated through each training set in this particular case:

$$fitness = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y_i})^2} \tag{3}$$

where y_i is the actual output value and $\widehat{y_i}$ is the predicted output value.

During each iteration, the velocity of each particle is updated towards their previous individual optimal position and global optimal position in a multidimensional search space, and then the particle position is updated according to the new velocity. After satisfying the predefined criterion, it jumps out of the iteration thus obtaining the optimal solution. The equations of particle update operations are as follows (Shi[13]):

$$V_{iD}^{j+1} = \omega V_{iD}^{j} + c_1 r_1 \left(p_{iD}^{j} - x_{iD}^{j}\right) + c_2 r_2 \left(p_{gD}^{j} - x_{iD}^{j}\right) \tag{4}$$

$$x_{iD}^{j+1} = x_{iD}^{j} + v_{iD}^{j+1} \tag{5}$$

where i stands for the particle and j denotes the current iteration number; D is the particle search dimension; x_{iD}^{j} and v_{iD}^{j} are the particle position and velocity in the j iteration; non-negative acceleration factor c_1 represents the individual's cognitive ability and c_2 represents the social guidance of the group; r_1 and r_2 are independently evenly distributed random numbers in the interval (0, 1); ω is the inertia weight which controls the impact of previous particle velocity on its current one in the solution domain, which is achieved by:

$$\omega = \omega_{max} - \frac{j}{j_{max}}(\omega_{max} - \omega_{min}) \tag{6}$$

in which j_{max} is the maximum iteration number. And this formula indicates that the inertia weight linearly decreases from ω_{max} to ω_{min} as the PSO technique progresses from the first iteration to the last one. It allows the algorithm to initially explore the search space in a wide range and then gradually focus more on fine-tuning the best-found solutions, hence achieving a balance between exploration and exploitation.

And the incorporation of PSO with the BP neural network offers several advantages that can address certain limitations of a traditional BP neural network. Primarily, PSO is a global optimization method, which can reduce the chance of the BP neural network getting stuck in local minima during the learning process. Also, PSO can facilitate a faster convergence of the learning process, and reduce the computational time required to train the neural network. Furthermore, by finding a more global solution, the PSO-BP method can reduce the risk of overfitting, which is also a common problem in the neural network training process. Lastly, PSO provides a more robust approach with few parameters to adjust and makes the BP neural network less dependent on the initial conditions since it will provide the global optimal values to BP.

The PSO-BP method's procedure can be expressed as follows with the overall prediction module also shown in Fig. 2:

(1) Define the architecture of the BP neural network from the dataset.
(2) Initialize weights and biases in the BP neural network.
(3) Initialize the particle velocity, position, individual optimal value and global optimal value in the PSO algorithm.
(4) Evaluate the particle fitness value individually according to the fitness function. The individual optimal value is updated if the current value is better. And the same approach is executed for the global optimal value if the individual optimal is better.
(5) Calculate the new particle velocity, position as well as perform the mutation operation.
(6) Return to step (4) if the current number of iterations is less than the maximum iteration value or the error parameter is larger than the set error value.
(7) Utilize the obtained optimal weights and biases from the PSO process to train the BP neural network.
(8) Calculate the output error through the loss function, propagate the error backward and adjust the weights and biases that minimize the error.
(9) Repeat step (8) until the network's prediction result meets certain criteria.

According to Feng[14], the equation $2i + 1$ provides an upper bound for the number of hidden neurons needed in a one-hidden-layer BP neural network, where i denotes the number of inputs. In this research, we utilized a 4-9-1 three-layer neural network structure for both the BP and PSO-BP methods. We examined a total of 18 combinations of PSO hyper-parameters as shown in Table 1 and the optimal set is highlighted in bold.

Thus, the PSO hyperparameters are set as follows: acceleration factors $c_1 = c_2 = 2.0$, population size is 30, evolutionary generation equals 50. The intervals of particle position and velocity are $[-5, 5]$ and $[-1, 1]$, ω_{max} and ω_{min} are set to be 0.9 and 0.4, respectively.

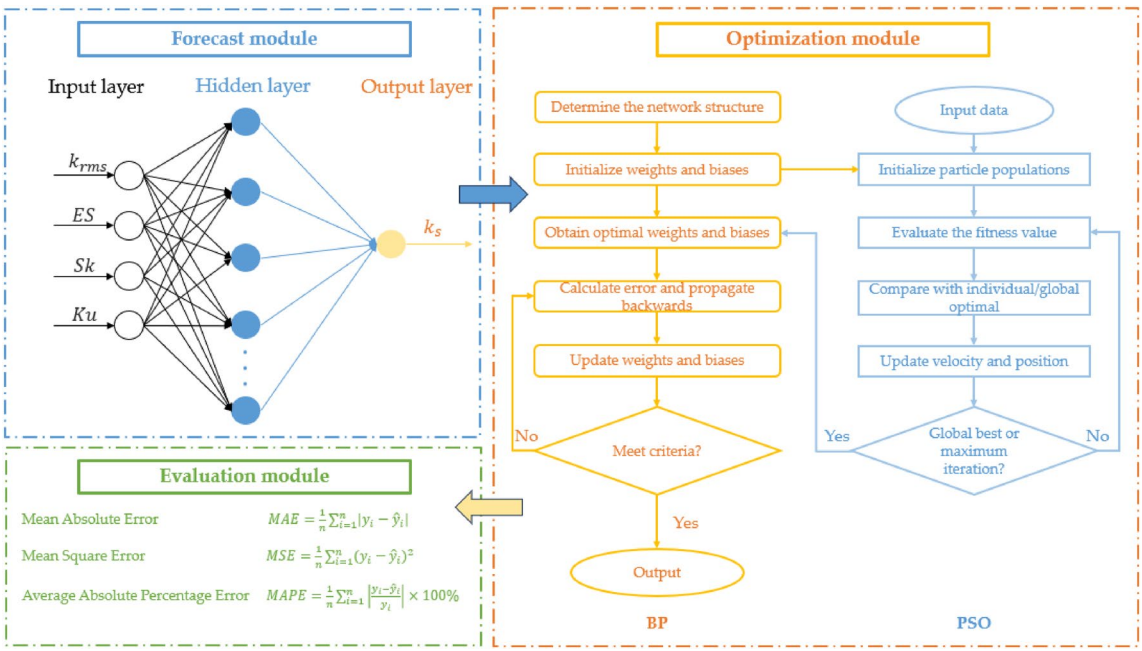

Figure 2. Overall prediction module of the PSO-BP algorithm.

Acceleration factors c_1, c_2	1.49445, **2.0**
Population size	**30**, 40, 50
Evolutionary generation	30, 40, **50**

Table 1. Range of hyper-parameters tested. Optimal values are in bold.

K-fold cross-validation

K-fold cross-validation is a resampling method used to evaluate machine learning models, especially useful for the current case where data samples are limited. It not only provides a more robust evaluation compared to a single train but also helps reduce overfitting. This procedure begins with the shuffling of original dataset, followed by its division into k subsets (or folds) of approximately equal size. Through each iteration, $k-1$ folds are used for training while the remaining one is allocated for testing. This process is repeated k times, ensuring each fold is used for test exactly once. After completing k iterations, k distinct models and their corresponding evaluation metrics are constructed, and their averaged results can provide a comprehensive assessment of the model's capabilities. In this study, we employed the fivefold cross-validation, and the optimal as well as averaged results are presented subsequently.

Model evaluation indicators

Evaluating the performance of the PSO-BP method is a crucial step, providing an understanding of the reliability of our predictions and aiding in the comparison of different models. In this study, we employed four key evaluation metrics: coefficient of determination (R^2), Mean Absolute Error (MAE), Mean Squared Error (MSE), and Mean Absolute Percentage Error (MAPE) to evaluate the existing roughness correlation formulas, the trained results from PSO-BP and traditional BP methods, as well as the optimized polynomial functions. And the expressions for these metrics are as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \overline{y_i})^2} \tag{7}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \widehat{y}_i\right| \tag{8}$$

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2 \tag{9}$$

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \widehat{y}_i}{y_i}\right| \times 100\% \tag{10}$$

among the aforementioned formulas, n is the total number of data points, y_i represents the actual value, $\widehat{y}_i$ denotes the corresponding prediction value and $\overline{y_i}$ is the mean of actual values.

Results

Data source and pre-processing

The database utilized for training the BP and PSO-BP neural networks as well as the roughness correlation formulas, consist of direct numerical simulation (DNS) results from Forooghi[5], Kuwata[7], Jouybari[8], Jelly[15], large eddy simulation (LES) results from Marchis[16], along with experimental results from Flack[6,17,18] and Womack[19], which is totally comprised of 86 distinct datasets and all of them reside within the fully rough regime. The roughness types cover a broad scale, including both regular and irregular, 2D and 3D roughness, ranging from ellipsoidal elements, sand-grain, turbine blade roughness, to grit blasting, truncated cone roughness, and so on. Here, k_{rms} and k_s values from DNS and LES results are all normalized by the channel half-height δ, while those values from Womack[19] are normalized by the boundary layer thickness. The ranges of the investigated parameters are as follows: $0.01 < k_{rms} < 0.12, 0.07 < ES < 1.14, -1.44 < Sk < 3.96, 1.43 < Ku \leq 26.8$. A subset of the sample data is presented in Table 2, while the complete dataset can be found in the Supplementary Materials.

Next, we conducted a feature importance analysis using the Random Forest model which gives an indication of how useful or valuable each feature is toward the model's predictive performance. This analysis not only supplements the insights gained from the previous heatmap, but also deepens our understanding of the complex interplay between those features especially for our dataset. It allows us to identify the most influential features among those four input variables, particularly when the dataset contains a variety of roughness types. Above all, this analysis guides the selection of appropriate roughness correlation formulas following the feature importance.

As illustrated in Fig. 3, k_{rms} emerges as the most influential feature, contributing significantly to the prediction of k_s. It is followed by Sk, ES and Ku, in descending order of importance. Thus, roughness correlation formulas prioritizing ES are set aside (e.g., Bons[20], Abdelaziz[21], Marchis[16]), and we opt for roughness correlation formulas from Flack[6] and Kuwata[7], in which the prediction of k_s rely heavily on k_{rms} and Sk. Besides, the roughness

k_{rms}	ES	Sk	Ku	k_s
0.0219	0.5949	− 0.1671	1.6007	0.0644
0.0326	0.9158	0.1093	1.4355	0.1244
0.0218	0.5876	− 0.0874	1.5898	0.0589
…	…	…	…	…
0.0280	0.3400	0.0100	3.0000	0.1439
0.0280	0.3400	0.0100	2.9800	0.1688
0.0310	0.3500	− 0.0200	2.9400	0.2062

Table 2. Partial sample data which are all in the fully rough regime.

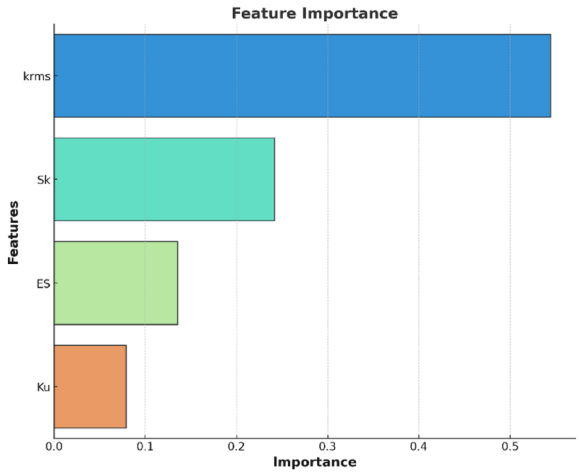

Figure 3. Feature importance plot measured by Random Forest.

correlation equation from Musker[22], which accounts for k_{rms}, Sk, ES and Ku simultaneously is considered. The selection of these correlation formulas aims to enhance the prediction accuracy for the given dataset, allowing for a more equitable comparison with neural networks in subsequent sections.

Prediction with established correction formulas

Firstly, we employed the roughness correlation formula proposed by Flack[6] with sample data, given as

$$k_s = 2.91k_{rms}(2 + Sk)^{-0.284} \tag{11}$$

and compare the predicted values with the actual ones. The obtained evaluation metrics to quantify the performance of this correlation formula are as follows: MAE is 0.1034, MSE is 0.0183, and MAPE is 51.66%.

Secondly, we utilized the formula developed by Kuwata[7]:

$$k_s = 4.0k_{rms}(1 + 0.17Sk)^4 \tag{12}$$

and in this case, the calculated MAE is 0.0729, MSE is 0.0141, and MAPE is 56.91%.

Lastly, Musker[22] 's equation has the form:

$$k_s = k_{rms}(1 + 0.5ES)(1 + 0.2Sk \cdot Ku) \tag{13}$$

where the calculated MAE is 0.1316, MSE is 0.0255, and MAPE is 76.22%.

Based on the evaluation metrics, Kuwata's roughness correlation formula demonstrates superior performance in terms of accuracy, evidenced by the lowest MAE and MSE values under the given dataset, though it shows relatively higher MAPE than Flack's result. One possible reason is that the rough surfaces generated by randomly distributed hemispheres in Kuwata's case make it more general and enhance its ability to capture varied surface characteristics. On the other hand, Flack's correlation formula is based on grit blasting and Musker's equation considers ship-hull roughness, such specificity might limit their adaptability compared to Kuwata's approach.

Prediction with BP neural network

The optimal prediction results using BP neural network are evaluated through the scatter plots of true k_s vs. predicted k_s values and scatter plot of relative error as well as the evaluation metrics (based on the test set):

and the calculated MAE is 0.0499, MSE is 0.0033 and MAPE is 31.84%. It's worth noting that in the k_s true vs. k_s predict values plot (left), the training and test sets overlap because the prediction results for both sets are close in those regions. On the other hand, the relative error plot (right)'s horizontal axis only represents sample size, and the test set is positioned after the training set for better visualization. As shown by Fig. 4a, most of the points follow the general trend of the regression line, while some deviations exist. Meanwhile, the relative error plot in Fig. 4b exhibits an overall symmetric pattern around zero, indicating the BP neural network's predictions are, on average, fairly accurate. But there are a few extreme values exist that represent outliers.

Prediction with PSO-BP neural network

The optimal prediction results using PSO-BP neural network are evaluated as follows: and the MAE, MSE and MAPE have been calculated as 0.0255, 0.0010, and 18.46%, respectively. As depicted in the left plot of Fig. 5, data points, in general, lie closer to the regression line than in the BPNN scatter plot. The PSO-BPNN yields more accurate prediction results especially for small k_s values, while larger k_s values present some discrepancy from the actual values. One potential explanation for this inconsistency may lie in the limited number of data points for large k_s values in the dataset. By incorporating additional samples representing larger k_s values, it might be possible to further increase the model's accuracy, thereby reducing the observed discrepancy. In addition, the overall relative error values are closer to zero which means the magnitude of errors is reduced, meanwhile, extreme values are alleviated.

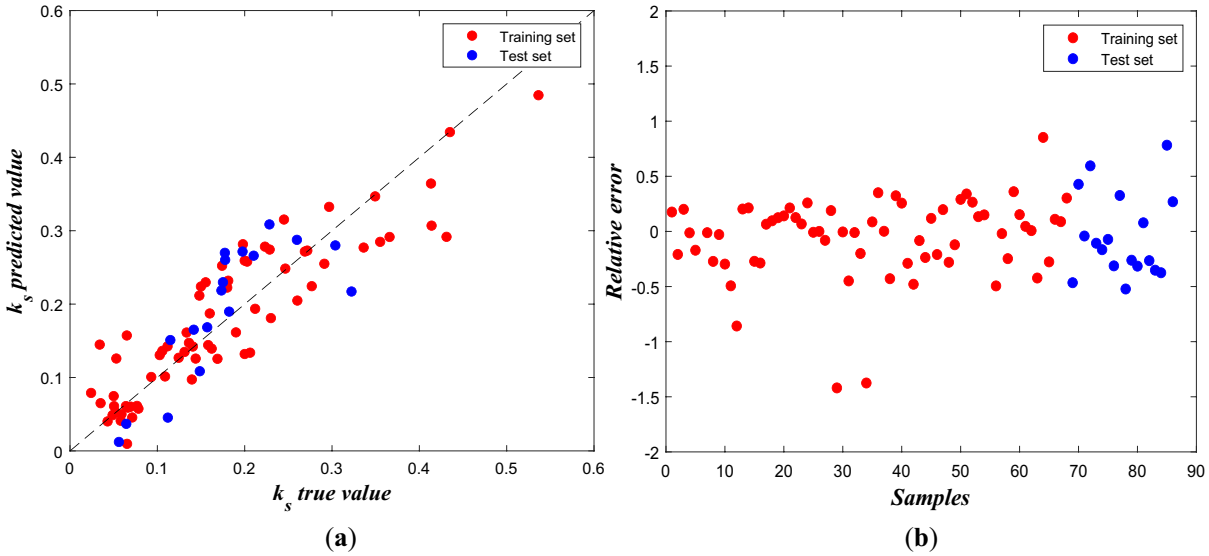

Figure 4. (**a**) Scatter plot of true k_s and predicted k_s, (**b**) Scatter plot of relative error for each sample.

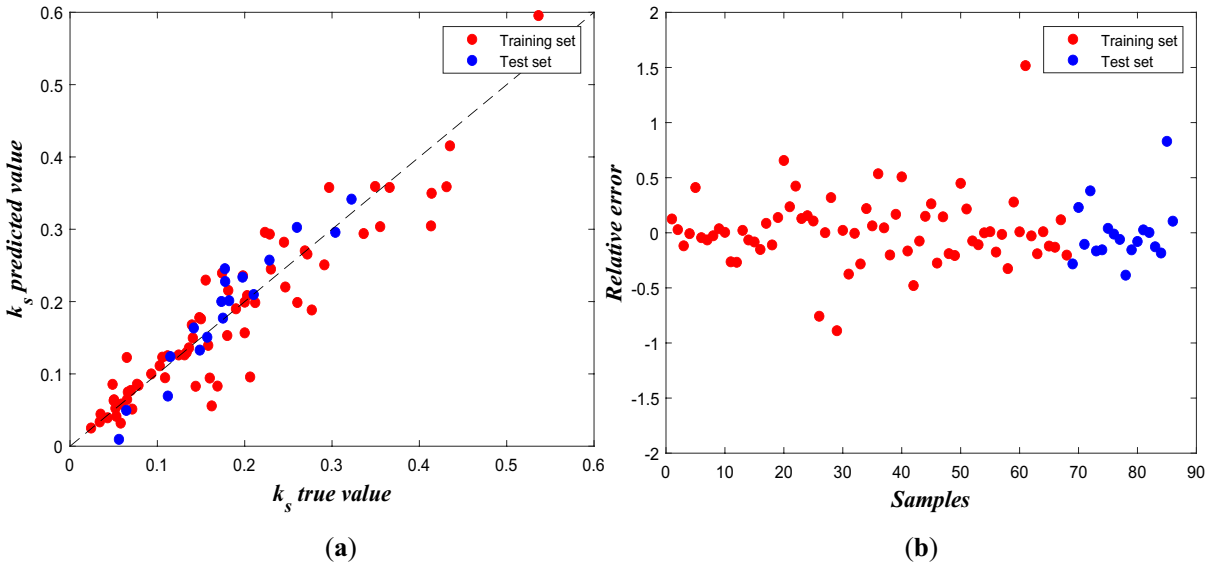

Figure 5. (**a**) Scatter plot of true k_s and predicted k_s, (**b**) Scatter plot of relative error for each sample.

The outcomes of the fivefold cross-validation process are presented in Table 3. To provide a more synthesized perspective, the averaged evaluation metrics of BPNN and PSO-BPNN are intuitively represented through bar plots in Fig. 6.

and the averaged MAPE for BPNN and PSO-BPNN are 46.07% and 28.12%, respectively. The comparison between BPNN and PSO-BPNN reveals critical insights into the efficacy of incorporating particle swarm optimization into BPNN. An obvious observation is the stark contrast in the averaged R^2 values. BPNN, although performing satisfactorily during the training set, exhibits a drastic deterioration in the test phase with an averaged R^2 value of 0.282. This decrease is notably attributed to the R^2 value falling below 0 in the 5th cross-validation of BPNN's test set, indicating the BPNN fails to capture the underlying data pattern. This could be due to the existence of outliers or extreme values in the test set, which are unrepresented in the training set, thereby skewing the model's predictive capability.

Conversely, PSO-BPNN demonstrates commendable robustness and generalizability. This is evident not just in the marginal reduction in averaged R^2 from training to testing, but also in its resilient R^2 value of 0.593 in the 5th cross-validation of the test set. Such consistency is indicative of the model's enhanced capacity to mitigate overfitting, likely attributable to the particle swarm optimization's role in efficiently navigating the solution space for optimal weights and biases. Moreover, PSO-BPNN consistently outperforms BPNN across other evaluation metrics as well, as evidenced by its lower MAE, MSE and MAPE in nearly every iteration, thereby implying a higher degree of accuracy and reliability.

Optimized polynomial model

While the 'black box' PSO-BPNN method exhibits satisfying results in predicting the equivalent sand-grain roughness k_s in the fully rough regime across variety of roughness types, the utilization of explicit algebraic representations, such as using polynomial functions to fit complex nonlinear models provides an alternate intuitive approach for the prediction of k_s. The appeal of this approach lies in its transparency and interpretability, and this is achieved through the curve-fitting procedure, where a basis function is first proposed for the model, and the unknown coefficients of the function are subsequently optimized under certain predefined constraints. In this study, an optimized polynomial function based on Musker[22] 's roughness correlation type has been proposed as a 'white box' model for the k_s prediction, which is straightforward and complementary to the previous 'black box' machine learning approaches. And the mathematical representation of the polynomial model is given by:

$$k_s = a_0 k_{rms}(a_1 + a_2 ES)^{a_3}(a_4 + a_5 Sk \cdot Ku) \tag{14}$$

where $a_i (i = 0, 1, \ldots, 5)$ are the coefficients to be determined for the polynomial function.

Here the curve-fit function from the optimize module in the Scipy library is adopted to find the optimized values of coefficients. It uses the Levenberg–Marquardt algorithm which is an iterative algorithm used for solving

	BPNN						PSO-BPNN					
	R^2		MAE		MSE		R^2		MAE		MSE	
No	Train	Test	Train	Test	Train	Test	Train	Test	Train	Test	Train	Test
1	0.816	0.319	0.0384	0.0499	0.0024	0.0033	0.853	0.794	0.0316	0.0255	0.0019	0.0010
2	0.619	0.055	0.0505	0.0738	0.0044	0.0095	0.831	0.883	0.0306	0.0280	0.0020	0.0012
3	0.496	0.515	0.0515	0.0701	0.0050	0.0076	0.791	0.753	0.0277	0.0521	0.0021	0.0039
4	0.759	0.700	0.0378	0.0474	0.0025	0.0045	0.903	0.713	0.0220	0.0488	0.0010	0.0043
5	0.870	-0.18	0.0319	0.0654	0.0016	0.0072	0.823	0.593	0.0337	0.0407	0.0021	0.0025
Average	0.712	0.282	0.0420	0.0613	0.0034	0.0064	0.840	0.747	0.0291	0.0390	0.0018	0.0026

Table 3. Statistics of cross-validation evaluation index.

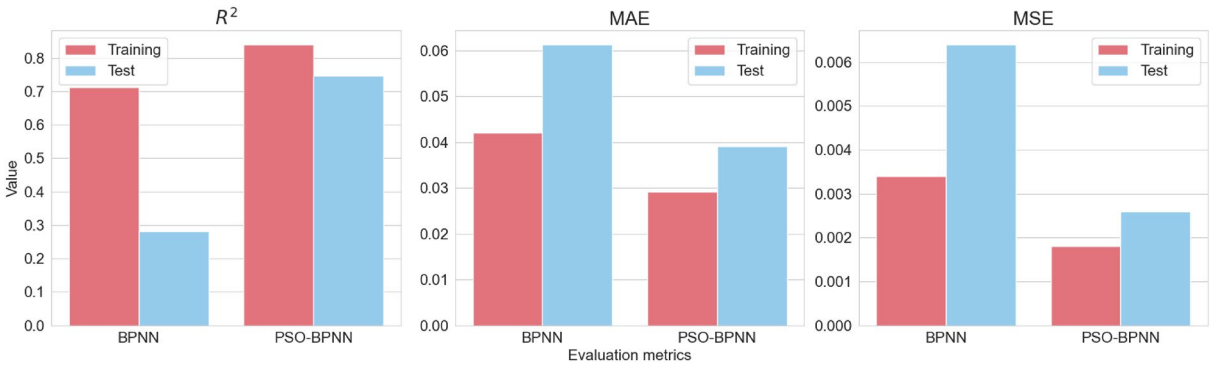

Figure 6. Bar plots of performance evaluation metrics.

nonlinear least square problems. This algorithm combines two optimization methods, namely, gradient descent and the Gauss–Newton method, aiming to provide robust and efficient converge performance under various conditions. The optimized coefficients are shown in Table 4:

and the prediction quality of the optimized polynomial function can be achieved through the evaluation metrics, resulting in a calculated MAE is 0.0608, MSE is 0.0074, and MAPE is 38.87%.

Then we can summarize and compare the performance of the preliminary roughness correlation formulas with the data-driven methods, including the optimized 'white box' model and PSO-BPNN, and the comparison is conducted based on the evaluation metrics of MAE, MSE and MAPE:

It can be seen from the evaluation metrics that the accuracy of this optimized explicit prediction model is relatively lower than that of the PSO-BP method, but is more promising than results from other selected roughness correlation formulas under the given dataset. Another thing to be noted is that MAPE values for roughness correlation formulas, optimized polynomial model and neural networks are relatively large for this particular problem, that's because k_s values in the dataset are close to zero which will be heavily penalized when calculating with the MAPE equation. Overall, it is apparent from Table 5 that the PSO-BPNN outperforms both the selected roughness correlation formulas and the optimized polynomial model among the evaluation metrics of MAE, MSE and MAPE. And it can be concluded that the PSO-BPNN model is a rather effective and accurate method to predict the equivalent sand-grain roughness k_s in the fully rough regime at various roughness types.

Uncertainty analysis

Recognizing and quantifying uncertainty is crucial for making informed decisions and understanding the reliability of PSO-BPNN's prediction toward k_s. One of the most effective methods to visualize and comprehend the uncertainty of model predictions is through Confidence Intervals (CI). Thus, by using previous results from the PSO-BP model and computing CI with the predicted k_s values, we obtain a probabilistic range wherein the true k_s value is likely to fall, with a specified level of confidence. This approach not only provides a quantitative measure of the model's uncertainty but also offers a visual tool to grasp the potential variations in predictions.

For our study, a 90% confidence level was chosen, which indicates that we are 90% confident that the true k_s value lies within the computed intervals. The confidence interval is calculated using the following formula:

$$CI = \overline{x} \pm t \times \frac{\sigma}{\sqrt{n}} \tag{15}$$

where $\overline{x}$ is the sample mean, t value is a critical factor determined by confidence level and degrees of freedom (approximately 2.13 in this case), σ denotes the sample standard deviation and n is the sample size. The results of this calculation are depicted in Fig. 7.

$a_0 = 0.8878$	$a_1 = -0.1987$	$a_2 = 2.5543$
$a_3 = 0.3437$	$a_4 = 5.0568$	$a_5 = 0.3626$

Table 4. Optimized values of a_i for the polynomial function.

	Musker et al.	Flack et al.	Kuwata et al.	'White box' model	PSO-BPNN
MAE	0.1316	0.1034	0.0729	0.0608	0.0390
MSE	0.0255	0.0183	0.0141	0.0074	0.0026
MAPE	76.22%	51.66%	56.91%	38.87%	28.12%

Table 5. k_s prediction statistics comparison.

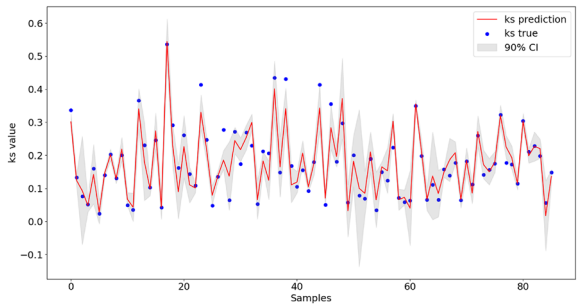

Figure 7. 90% Confidence Interval for PSO-BPNN.

From the confidence interval plot, it's evident that most of the k_s true values fall within the 90% confidence interval, suggesting that the PSO-BPNN model is generally accurate and reliable for most predictions. And the model's ability to follow the general trend of the k_s true values indicate a good understanding of the underlying data patterns and relationships between surface parameters and k_s. However, regions with noticeably broader confidence intervals highlight where the model exhibits higher uncertainty. This increased uncertainty may be attributed to outliers or regions in the input space where the model is less trained. Yet, larger k_s values tend to deviate from the confidence interval, aligning with the observation before that the PSO-BP model has less prediction accuracy for large k_s.

While the PSO-BP model demonstrates commendable robustness, areas of heightened uncertainty and deviations in larger k_s values underscore potential avenues for improvement. It's important for future work to probe deeper into these regions of increased uncertainty and train with more large k_s values, enhancing the model's comprehensiveness and predictive accuracy across the entire problem domain.

Discussion

The PSO-BPNN model results reveal its efficiency in predicting the equivalent sand-grain roughness k_s and demonstrate that neural networks can be successfully utilized for this kind of prediction task. In addition, the implementation of PSO in conjunction with BP significantly improves the prediction results, as evidenced by the reduction in MAE, MSE and MAPE by 36.38%, 59.38% and 38.96% respectively, when compared to the BP model alone. The PSO algorithm helps to optimize the weights and biases in the BP neural network, and this process can be considered as an implicit form of feature selection which enhances the model's ability to accurately represent the dataset. This finding substantiates the effectiveness of PSO in optimizing of BP, underscoring the PSO-BP model as a more efficient method for the prediction of k_s.

In fact, the development of a predictive model for k_s with arbitrary roughness types, based on multiple surface topography from a selected dataset, is labeled a complex nonlinear regression problem that is well-suited to the ML technique. Since various surface roughness yield different flow phenomena and makes a general physical model difficult to achieve, the ML technique takes no account of the physical modeling concept or observed phenomena within roughness sublayers for each case, nor is it affected by the lack of orthogonality among surface parameters chosen as the independent variables toward k_s. The flexibility of ML enables it to mimic models with many degrees of freedom or multivariate problems, and its feature selection property allows for the calibration of polynomial coefficients across different regions of the surface parameter space, while in a more thoroughly manner. In contrast, the reduced accuracy in roughness correlation formulas and optimized polynomial model can be attributed to the limitations inherent to low-order polynomial functions, which may not adequately capture the dependence of k_s on surface parameters. And this is due to the constraint that each coefficient in the polynomial function must maintain a uniform value across the entire surface-parameter space. Even though, the optimized 'white box' polynomial model, which balances the trade-off between predictive accuracy and transparency, could serve as a valuable baseline for future studies aiming to parameterize k_s explicitly across a broad spectrum of roughness types.

Conclusion

In this study, we first utilized datasets from the fully rough region to devise the predictions for k_s according to the analyzed most important surface parameters, namely the root-mean-square roughness height k_{rms}, effective slope *ES*, skewness *Sk* and kurtosis *Ku* with both BP and PSO-BP methods. Next, a Random Forest analysis is performed to find the appropriate roughness correlation formulas corresponding to the feature importance result. It also reveals that k_{rms} emerges as the most influential feature toward the parameterization of k_s, followed sequentially by *Sk*, *ES* and *Ku* in descending order of importance. In addition, a 'white box' polynomial function for k_s prediction is proposed based on the curve-fit technique. We also conducted an uncertainty analysis to assess the reliability of the PSO-BP model's predictions. It turns out the PSO-BP method offers a better accuracy among the existing roughness correlation formulas, the traditional BP method as well as the explicit optimized polynomial functions. Furthermore, the training time for the PSO-BP method is less than 2 min with few hyperparameters to tune, making it a time-saving and efficient ML approach.

We therefore conclude that the PSO-BP method yields high-fidelity predictions of k_s for turbulent flows over a wide range of rough surfaces and outperforms other listed methods, hence providing efficient and reliable inputs for further roughness-related CFD simulations or other engineering applications. In order to improve the model accuracy in further study, more data sets containing surface parameters and corresponding equivalent sand-grain roughness, especially larger k_s values are needed to train the model, and enhancement could also be achieved by incorporating additional roughness parameters as input features, such as solidity, porosity and so on.

Data availability

The dataset examined in the current study is presented in this paper and supplementary materials.

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Author contributions

Conceptualization, H.M.; methodology, H.M. and X.Y.; software, X.Y.; validation, H.M., Y.L. and X.Y.; formal analysis, H.M.; investigation, Y.L.; data curation, H.M.; writing—original draft preparation, H.M, Y.L..; writing—review and editing, X.Y.; visualization, L.Y. All authors reviewed the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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